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Research Paper

YOLO26-EGH:A lightweight knowledge-distilled detection model for blueberry ripeness grading in greenhouse

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ABSTRACT

Improper blueberry maturity discrimination easily causes pre-harvest fruit losses, which severely restricts the high-quality development and economic benefits of the blueberry industry. Accurate and efficient maturity detection is critical for precision harvesting and loss reduction. Nevertheless, dense fruit occlusion, subtle color differences between adjacent maturity stages, and limited computing resources of edge devices make it difficult for existing models to balance detection accuracy and real-time deployment performance. To address these challenges, this paper proposes YOLO26-EGH, a lightweight blueberry maturity-detection model based on knowledge distillation. A self-built image dataset containing 3000 valid images was constructed across two acquisition years for two blueberry cultivars, F6 and F11. Built upon YOLO26n, the C2PSA-EMA module is introduced to enhance feature representation of weak color differences and local details; the C3k2-GCB module is adopted to strengthen multi-scale context modeling; and HSFPN is utilized to optimize multi-layer feature fusion while lightweighting the network. Offline knowledge distillation is further implemented to boost the detection performance of the lightweight student model. Results of repeated comparative experiments and two-tailed paired-samples t-tests demonstrate that, compared with the YOLO26 baseline, YOLO26-EGH achieves statistically significant improvements in Recall, mAP@50 and mAP@50:95 while maintaining stable Precision. The proposed model obtains a Precision of 87.9%, a Recall of 87.9%, mAP@50 of 93.2% and mAP@50:95 of 82.3%. Competitive performance is also achieved on a public blueberry dataset, and embedded-device tests yield a real-time detection latency of 12.72 ms. This method can provide effective technical support for intelligent blueberry maturity monitoring and precision-harvesting operations.

1. Introduction

According to the industrial annual report of the Small Berry Branch of the Chinese Society for Horticultural Science, China's total blueberry output reached 780,000 tons in 2024. With the large-scale development of protected cultivation, blueberries have gradually shifted from high-end niche fruits to popular fresh produce (Li et al., 2025a; Xu et al., 2024). Blueberries have a narrow optimal harvesting window, and relying on manual experience to judge maturity often leads to delayed picking. Over-ripeness, softening, and rot are the main causes of pre-harvest loss in berries. Improper maturity control accounts for the majority of losses across the entire industrial chain, resulting in substantial economic losses each year (Wang et al., 2025; Godara et al., 2025; Fuentes et al., 2026). Automatic and accurate recognition of blueberry maturity using machine vision technology can reduce over-ripe fruit loss from the harvesting source, serving as a key approach to improving blueberry commodity rate and reducing postharvest losses (Wu et al., 2025; Ni et al., 2021).

Deep learning-based object detection serves as a core technique in machine vision, which can be broadly categorized into two-stage, single-stage, and Transformer-based detection paradigms. Two-stage detection algorithms represented by Faster R-CNN first generate region proposal candidates and subsequently perform object classification and bounding box regression, achieving superior localization accuracy while suffering from complicated inference pipelines and high computational costs (Ren et al., 2017; Varghese and Sambath, 2023). In contrast, single-stage detectors including SSD, RetinaNet, and YOLO directly predict object categories and bounding coordinates, establishing an effective trade-off between detection accuracy and inference efficiency (Hmidani and Alaoui, 2025). Transformer-based detection models, represented by the DETR series, integrate global attention mechanisms and end-to-end prediction paradigms into object detection tasks, substantially

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enhancing the modeling capability for complex spatial relationships and multi-scale feature representation (Zhu et al., 2021; Zhao et al., 2024). Nevertheless, most Transformer-based models suffer from massive computational overhead and high memory consumption. Accordingly, lightweight single-stage detection models are more suitable for real-time detection tasks in greenhouse and orchard scenarios, and have been widely adopted in blueberry detection research.

Blueberries are mainly divided into wild lowbush blueberry (Vaccinium angustifolium Ait.) and cultivated highbush blueberry (Vaccinium corymbosum). Wild lowbush blueberry plants are short in stature with tiny, sparsely distributed fruits, resulting in complex field environments covered by dense branches, leaves, and weeds. Targeting this challenging scenario, Mullins et al. (2024) adopted the zero-shot detection frameworks of YOLO-World and Grounding DINO to identify wild blueberries, weeds, flower buds, and diseases, verifying the feasibility of these methods for improving the labeling efficiency of wild blueberry datasets. MacEachern et al. (2023) compared six lightweight models of the YOLOv3 and YOLOv4 series to realize binary and ternary maturity classification of wild blueberries, and further constructed a regression model based on detection results to achieve field yield estimation. In comparison, existing research on blueberry maturity detection predominantly focuses on cultivated highbush blueberries. Li et al.(2025) constructed an in-field phenotyping robot platform for blueberry detection, integrating the SAM segmentation model with a self-developed lightweight BerryNet. Using multi-view imaging and a BiFPN feature fusion structure, the system achieved automatic intelligent detection of blueberry cluster segmentation, maturity classification, and cluster traits in field conditions. Fu et al. (2026) proposed a lightweight detection network DCS-YOLO to address the complex orchard scenarios with varying blueberry sizes and heavy occlusions. The model significantly reduced parameters and computational cost while accurately classifying blueberries into unripe, half-ripe, and ripe stages. Zhao et al. (2025) combined UAV-based vision with super-resolution reconstruction and a Mamba semantic segmentation network to realize high-precision segmentation and monitoring of blueberry maturity in open fields, effectively solving the problems of low efficiency and high cost in manual detection. Li et al. (2026) improved the lightweight YOLOv7tiny network into YOLOv7tiny-DGS to tackle low manual harvesting efficiency and poor recognition accuracy. The model enhanced maturity detection performance through multi-module feature optimization and was developed into a visualization system suitable for field regional maturity detection in orchards.

Current blueberry visual detection techniques have achieved substantial progress in detection accuracy, environmental adaptability, and lightweight optimization. Agricultural vision research has gradually evolved from single-target recognition toward multi-dimensional intelligent analysis integrating detection, segmentation, counting, and spatial distribution modeling (Cui et al., 2025). In general crop detection tasks, structural improvements such as attention enhancement and multi-scale feature optimization can effectively improve the detection stability of weakly featured and scale-varying objects in complex field environments (Wang et al., 2026a). Lightweight deployment strategies combining quantization-aware training, semi-supervised data optimization, and multi-scenario validation also provide feasible solutions for the practical application of agricultural detection models (Wang et al., 2026b). Nevertheless, existing lightweight blueberry detection models, such as BMDNet-YOLO, only achieve coarse three-stage maturity classification through structural optimization and transfer learning. These methods fail to capture subtle color and texture differences among blueberry fruits and cannot meet the requirements of fine-grained maturity identification (Sun and Wang, 2025). Model pruning and knowledge distillation have emerged as critical techniques for the lightweight deployment of detection models and have been widely applied in various fruit and vegetable detection tasks. Shi et al. (2025) combined dynamic channel pruning and channel-wise knowledge distillation to compress a multi-scale feature detection network, effectively alleviating feature coupling in dense object detection. Chen et al. (2024a) integrated pruning and block correlation knowledge distillation based on YOLOv8s, realizing model compression and accuracy recovery for maturity detection of colored melons. Lv et al. (2024) developed LDC-PP-YOLOE using localized knowledge distillation and CBAM optimization, which exhibited favorable lightweight performance and generalization in citrus detection tasks. Wang et al. (2025) achieved lightweight YOLOv8 via channel pruning, making it suitable for real-time detection on resource-constrained UAV platforms.

To address the problems of high redundancy and slow edge-side inference speed in existing blueberry maturity detection models, as well as the lack of customized optimization of distillation and pruning strategies for the field characteristics of blueberries such as dense occlusions and clustered small targets, this study proposes a lightweight detection model YOLO26-EGH for blueberry maturity classification. Built on the YOLO26 framework with its efficient feature extraction and inference speed advantages, the model incorporates an Efficient Multi-Scale Attention (EMA) mechanism (Ouyang et al., 2023) to improve the C2PSA module for enhanced fine-grained feature representation. The Global Context Block (GCBlock) (Cao et al., 2019) is adopted to optimize the C3k2 module for strengthened multi-scale blueberry cluster feature extraction. A lightweight High-level Screening-Feature Pyramid Network (HSFPN) (Chen et al., 2024b) replaces the original feature pyramid network to enhance multi-scale feature fusion while reducing model parameters. On this basis, an offline knowledge distillation strategy (Zhang and Ma, 2023) is introduced to further improve the detection performance of the lightweight model. The model is trained and validated on a two-year field dataset containing two major blueberry cultivars, F6 and F11, which ensures stable and generalized detection performance in real orchard scenarios. The proposed method can provide technical support for real-time blueberry maturity monitoring, harvesting-timing decision-making and lightweight edge deployment.

2. Materials and methods

2.1. Self-Constructed blueberry ripeness grading dataset

The first phase of blueberry image data in this study was collected from December 2024 to February 2025 at the Qingcaomu Modern Greenhouse Park in Jiancaoping District, Taiyuan City, Shanxi Province, containing only the single F6 blueberry cultivar, as shown in Fig. 1. To improve the multi-cultivar generalization ability of the model, supplementary images of the F11 blueberry cultivar were collected at the same park from December 2025 to February 2026. The two-phase data were merged to construct a dual-cultivar blueberry field dataset. The blueberry plants in the experimental area were cultivated in greenhouse pots with substrate, at a plant spacing of 50 cm and row spacing of 150 cm. Only healthy plants free of diseases, pests, and mechanical damage were selected as shooting targets to ensure sample consistency. A fully consistent imaging protocol was adopted throughout both phases of image collection to eliminate interference from acquisition variables on model training.

A unified imaging standard was applied in both sampling phases to avoid experimental variations. Sampling was conducted during stable light periods from 9:00–11:00 and 14:00–16:00 every day, starting 50 days after full bloom, with a sampling interval of 10 days. Three shooting distances were set: 30 cm, 45 cm, and 60 cm. Considering the greenhouse layout and the actual operating conditions of unmanned inspection vehicles, the maximum shooting distance was set at 60 cm to effectively avoid physical collisions between the inspection equipment and plants, ensuring safe field operation. Images under both frontlight and backlight natural illumination were collected synchronously, with a raw image resolution of 4096 × 3072 pixels. This setup preserved real-world field disturbances such as leaf occlusion and complex backgrounds in the greenhouse, meeting the data requirements for training

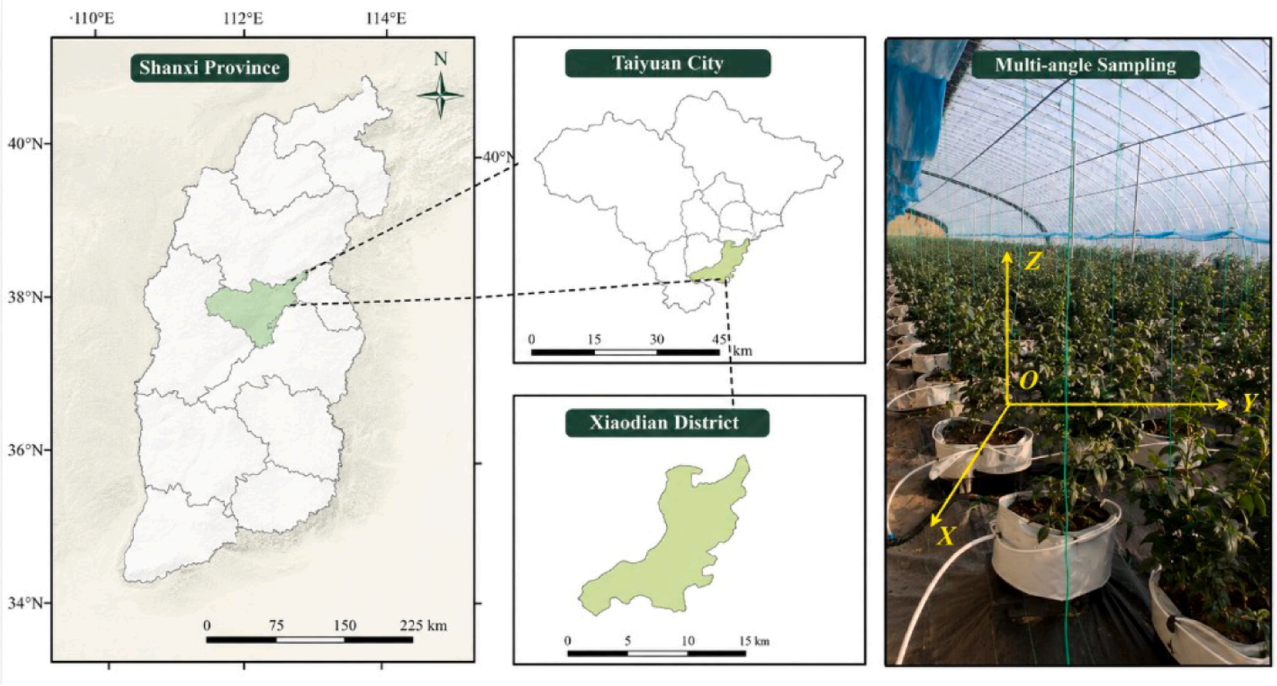

Fig. 1. Location of the blueberry experimental greenhouse and schematic illustration of the multi-angle image acquisition protocol.

and testing object detection models.

After removing invalid images such as blurred, overexposed, and severely occluded samples, a total of 3000 valid blueberry images were selected. The experiment focused on two key growth stages: the late expansion stage to the full ripening stage of blueberry fruits. Based on the biological development rules and morphological characteristics of the fruits, including peel color change, fruit appearance, and calyx retention status, the fruits were divided into five maturity grades: a, b, c, d, and e (Shi et al., 2023). Grade a corresponds to the early coloring stage, with the peel generally turquoise and no anthocyanin

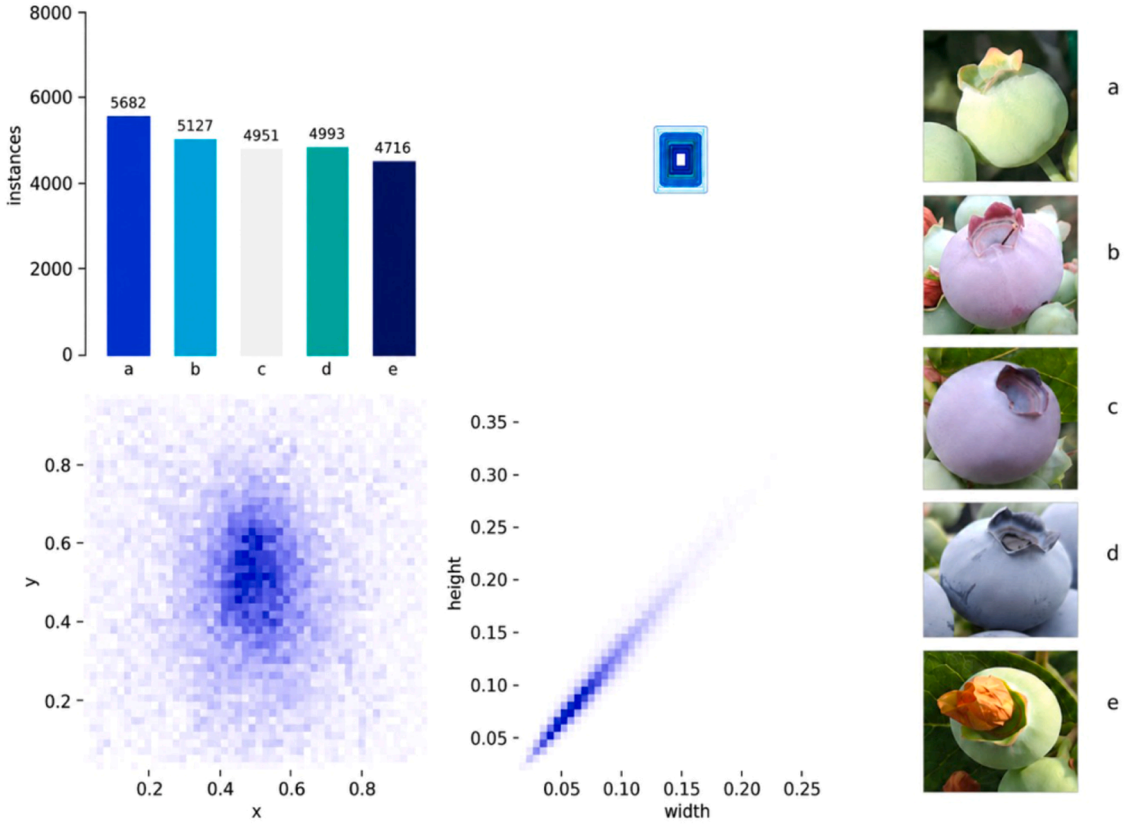

Fig. 2. Distribution of blueberry ripeness labels and annotated bounding-box characteristics.

accumulation. Grade b corresponds to the early-middle coloring stage, with the peel gradually turning pink as anthocyanin synthesis begins. Grade c corresponds to the middle-late coloring stage, with a large purple area on the peel and mid-stage coloring. Grade d corresponds to the full ripening stage, with an even dark blue peel and fully matured fruit development. Grade e corresponds to the late expansion stage, where fruit volume ceases to expand, the peel is smooth and uncolored, and persistent calyxes remain at the top.

In this study, the LabelImg tool was used for YOLO-format image annotation. The annotation rule was to label each fully visible blueberry and each fruit with slight leaf occlusion individually, while excluding severely occluded or morphologically incomplete fruits whose maturity could not be identified. The distribution of annotation labels is shown in Fig. 2. After image annotation, all samples were randomly shuffled and split into training, validation, and test sets at a ratio of 8:1:1.

2.2. External datasets for validation

To comprehensively evaluate the generalization ability of the improved model, a public blueberry field dataset was adopted for external validation. This dataset consists of multi-view blueberry images captured from front, side and oblique top perspectives, which contains field spatial and timestamp information as well as accurate fruit bounding box annotations. It supports blueberry fruit counting and spatial distribution detection tasks well. Limited by its image collection period, this dataset only covers blueberry samples in the coloring stage and excludes fruits at the late expansion stage. In accordance with the unified maturity classification criteria proposed in this study, the label corresponding to the late expansion stage was removed, and only four maturity categories of coloring-stage blueberries were applied for subsequent validation experiments. More detailed information about this public dataset is available in Li et al.(2023) and Li et al.(2025b).

2.3. Methods

Aiming at the defects of insufficient extraction of subtle color difference features, poor multi-scale feature fusion performance and limited detection accuracy of lightweight models of the original YOLO26 network in complex orchard environments, this study proposes an improved YOLO26-EGH detection model, as shown in Fig. 3. Four targeted improvement strategies are adopted in sequence. Firstly, the C2PSA-EMA attention module is added to enhance the middle-layer fine-grained feature representation and focus on effective feature information of blueberry targets. Secondly, GCBlock is introduced to reconstruct the C3k2 modules at P4 and P5 layers of the backbone network, so as to strengthen the low-level multi-scale feature extraction capability. Thirdly, the lightweight HSFPN replaces the original feature pyramid structure to optimize neck multi-scale feature fusion and reduce network parameters. Finally, offline knowledge distillation is adopted to transfer prior knowledge from large-scale models, further improving the detection accuracy of lightweight network variants.

2.3.1. C2PSA-EMA

Blueberry detection in orchard scenarios is susceptible to branch and leaf occlusion, dense fruit stacking and complex background interference. Meanwhile, blueberries at different maturity stages have extremely subtle differences in pericarp color. The original C2PSA module embedded in YOLO26 network has limited feature focusing capability, which fails to accurately extract fine-grained color features of blueberries and results in maturity misjudgment and small target missing detection. To solve this problem, this study introduces the efficient multi-scale EMA attention mechanism to replace the original attention unit of C2PSA, and constructs a lightweight C2PSA-EMA feature enhancement module (Ma et al., 2026). With a slight increase in computational overhead, the proposed module enhances the network's feature extraction capability for blueberry targets, suppresses background interference, and improves the maturity detection accuracy

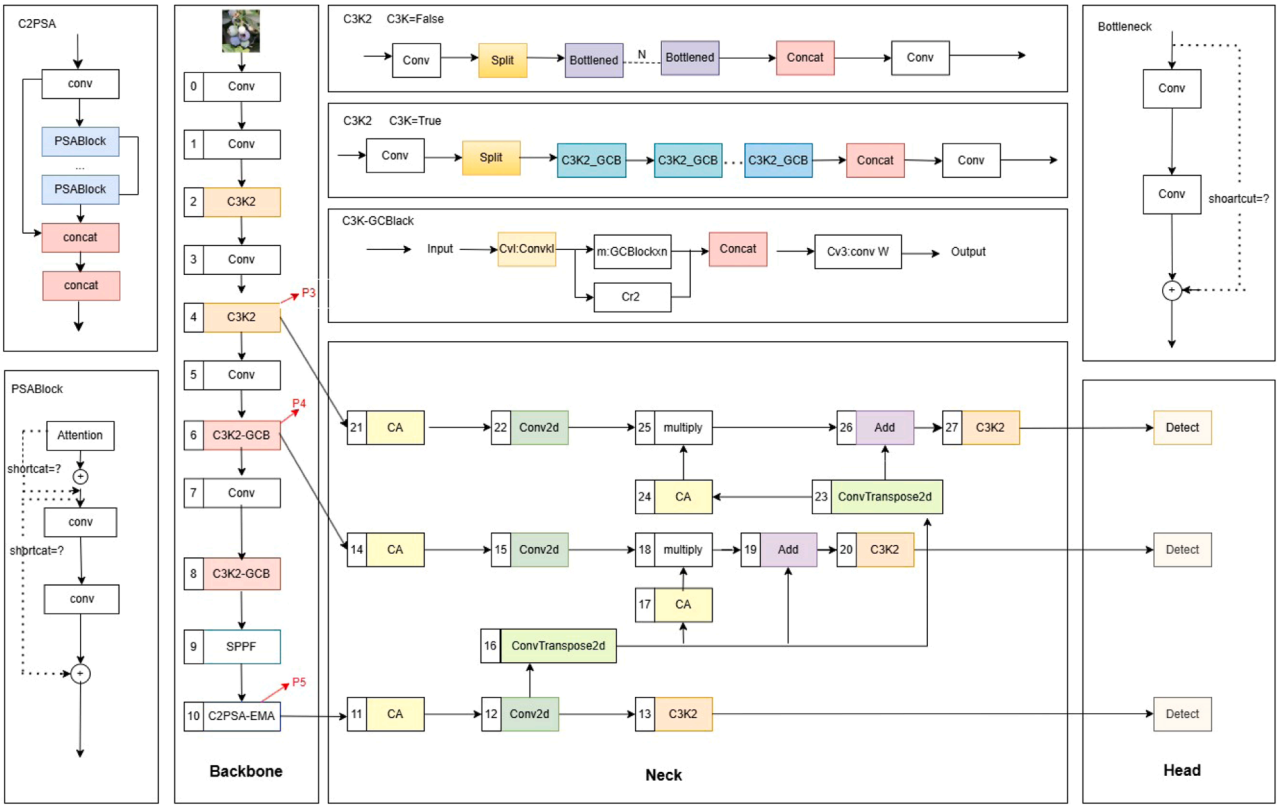

Fig. 3. Architecture of the proposed YOLO26-based blueberry ripeness detection network.

of blueberries in complex field environments.

The EMA attention mechanism takes channel grouping calculation as its core principle, and the grouping factor is fixed at 16 in this study. The input features are grouped along the channel dimension for parallel calculation. Horizontal and vertical adaptive average pooling are adopted to obtain global spatial dependencies in two directions. After being fused by 1 × 1 convolution, bidirectional spatial features generate spatial attention weights via the Sigmoid function to realize initial feature enhancement. Furthermore, a 3 × 3 convolutional branch is added to extract local texture and subtle color difference features of blueberries. After global average pooling and Softmax normalization, two groups of features integrate global spatial information and local detail information through matrix operation, and generate adaptive weights to optimize original features for the second time. The structure is shown in Fig. 4.

Based on the prominent feature enhancement performance of EMA, structural reconstruction and optimization are conducted on the original C2PSA module. Firstly, the C2PSA-EMA module doubles the channels of input features by 1 × 1 convolution, and then divides the processed features into two branches equally. One branch retains original shallow features directly, while the other branch is sent to cascaded PSABlocks for deep feature optimization. The built-in attention structure of each PSABlock is replaced by EMA, and two 1 × 1 convolutional layers are adopted to form a feedforward network. The number of adaptive attention heads is defined as the number of feature channels divided by 64, and residual connections are introduced to alleviate the gradient vanishing problem of deep networks. Finally, two branches are concatenated, and the original channel dimension is restored by 1 × 1 convolution to complete multi-scale feature fusion, which is well adapted to the blueberry maturity detection task based on YOLO26 network.

2.3.2. C3k2-GCB

In blueberry maturity detection tasks, low-level features mainly include basic information such as fruit edges, textures, and color gradients, whose extraction quality directly affects the representation performance of high-level features. The C3k2 module in the backbone of the original YOLO26 network adopts a conventional bottleneck structure, which has a limited perception range for fine textures and weak color difference features of blueberries, and tends to lose key low-level information under complex field backgrounds. Therefore, this study introduces the GCBlock and embeds it into the C3k2 modules at the P4 and P5 layers of the backbone network to form the C3k2-GCBlock structure, aiming to strengthen multi-receptive-field feature extraction at low levels and provide richer and more complete basic features for the subsequent C2PSA-EMA module.

GCBlock is a lightweight reparameterized convolutional module with a parallel multi-branch structure, consisting of two 3 × 3 convolutional branches, one 1 × 1 convolutional branch, and a residual branch (Tang et al., 2025). During training, multiple branches independently learn features at different scales; during inference, they can be fused into a single 3 × 3 convolution, expanding the receptive field and enhancing detail capture without increasing inference latency. Through complementary multi-receptive-field learning, the module effectively extracts texture, contour, and color difference features of blueberry fruits, and relies on residual connections to ensure stable feature transmission in low-level networks and avoid degradation of shallow information. The structure is shown in Fig. 5.

This study replaces the original convolutional units in the C3k2 module of the backbone network with GCBlock to construct the improved C3k2-GCBlock module. Following the dual-branch structure of C2f, the module splits feature channels via 1 × 1 convolution, with one branch retaining direct feature connection and the other performing multi-receptive-field feature extraction through stacked GCBocks. By enhancing low-level feature representation at shallow positions of the backbone, the module expands the effective receptive field, suppresses background noise interference, and supplies higher-quality input features for the subsequent C2PSA-EMA attention module, enabling the overall network to achieve stronger robustness and accuracy in recognizing subtle maturity differences of blueberries.

2.3.3. HSFPN

To streamline the network structure and reduce total parameters and computational cost while maintaining favorable multi-scale feature fusion performance, this study adopts the hierarchical selective feature pyramid network (HSFPN), the structure is shown in Fig. 6. It can effectively connect optimized features extracted by the backbone C3k2-GCBlock module and enhanced features generated by the middle-layer C2PSA-EMA module. As a lightweight feature fusion structure equipped with embedded attention units, HSFPN realizes adaptive feature purification. It reduces redundant network parameters without sacrificing detection accuracy, and meets the requirements of real-time inference and lightweight deployment for blueberry maturity detection.

HSFPN consists of two core components: channel attention (CA) module and feature selection module (Li et al., 2024). The CA module adopts parallel average pooling and max pooling branches to integrate global feature information, and adaptively calibrates channel weights to highlight critical feature channels related to blueberry color difference and texture. The feature selection module generates spatial attention masks via global average pooling and convolutional layers to enhance target features. Combined with residual connection, it retains original effective feature information, suppresses field background interference and prevents the loss of fine-grained feature details.

HSFPN matches the overall architecture of the improved YOLO26 network perfectly. Deployed at the neck part of the network, it receives low-level multi-receptive-field features from C3k2-GCBlock and middle-layer fine-grained enhanced features from C2PSA-EMA, and achieves efficient complementary fusion of high-level and low-level features. Compared with the direct feature concatenation adopted by the original FPN, HSFPN refines feature information from both channel and spatial dimensions. It establishes a complete optimization pipeline including underlying feature extraction, middle-layer feature enhancement and top-layer feature fusion, and further improves the comprehensive detection performance of the model in complex orchard scenarios.

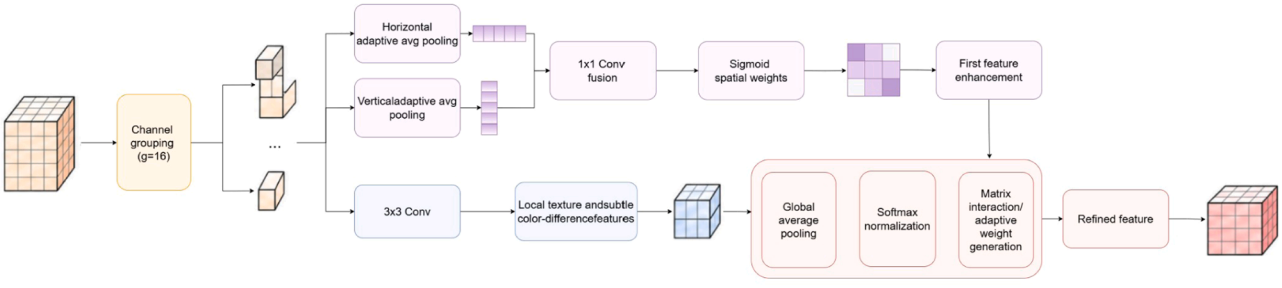

Fig. 4. Structure of the proposed C2PSA-EMA feature enhancement module.

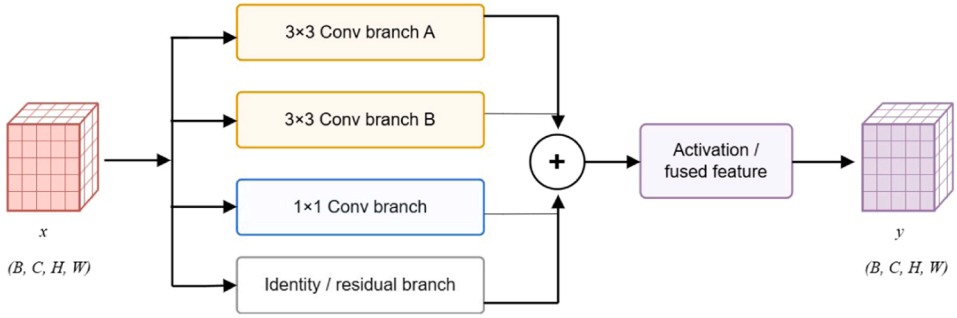

Fig. 5. Structure of the proposed C3k2-GCB module.

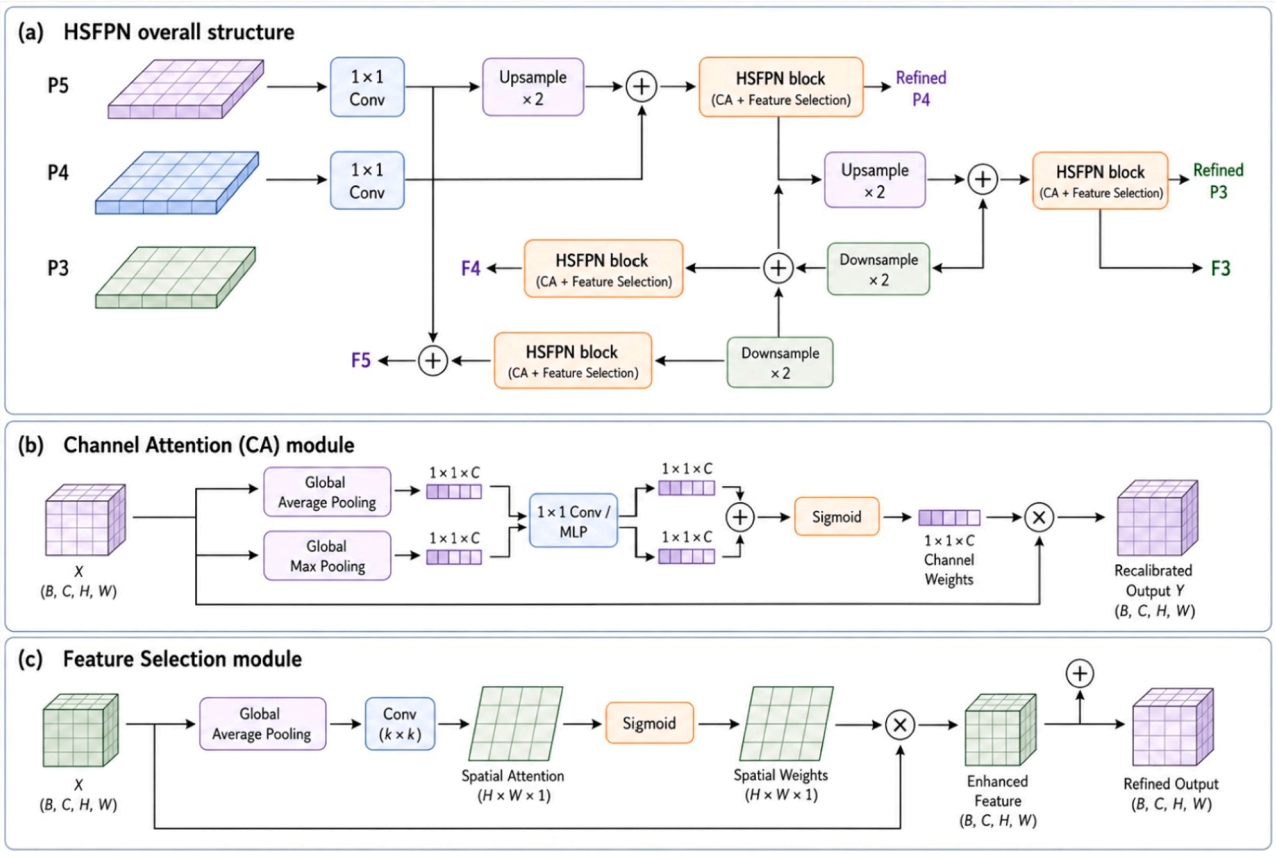

Fig. 6. Structure of the proposed HSFPN neck module.

2.3.4. *Distillation*

The original YOLO26 series consists of five models with different scales. Model parameters and detection performance decline with the reduction of network size. The large-scale YOLO26x has deeper network layers and sufficient convolution channels, thus possessing stronger feature extraction capability and inherent higher detection accuracy. In contrast, the lightweight YOLO26n has a compact structure and faster inference speed, yet it suffers from limited feature mining capacity and unsatisfactory detection accuracy due to its small network scale. To balance detection accuracy, inference speed and lightweight requirements, this study adopts offline knowledge distillation. The high-precision YOLO26x is taken as the teacher model to transfer implicit detection knowledge to the lightweight YOLO26n student model, so as to improve the accuracy of the small model without modifying its network structure.

Offline knowledge distillation is a two-stage model optimization method different from online distillation that requires synchronous training of teacher and student models. It includes two separate training phases (Zhang and Ma, 2023). Firstly, the teacher model is trained independently and its parameters are fixed. The well-trained teacher model generates soft labels containing implicit category correlation information. Compared with one-hot hard labels from original datasets, soft labels carry latent feature distribution and category difference knowledge learned by the teacher model. By minimizing the prediction error between teacher and student models, the student model can learn the decision-making logic and feature extraction rules of the teacher model, thereby improving the performance of the lightweight model efficiently. The structure is shown in Fig. 7.

In this work, YOLO26x and YOLO26n are selected as the teacher model and student model respectively for offline distillation. Firstly, all improved modules are integrated into YOLO26x, and the trained teacher model is frozen after independent training. During the training process of the student model, the frozen teacher model is loaded directly and conducts real-time forward propagation to provide distillation

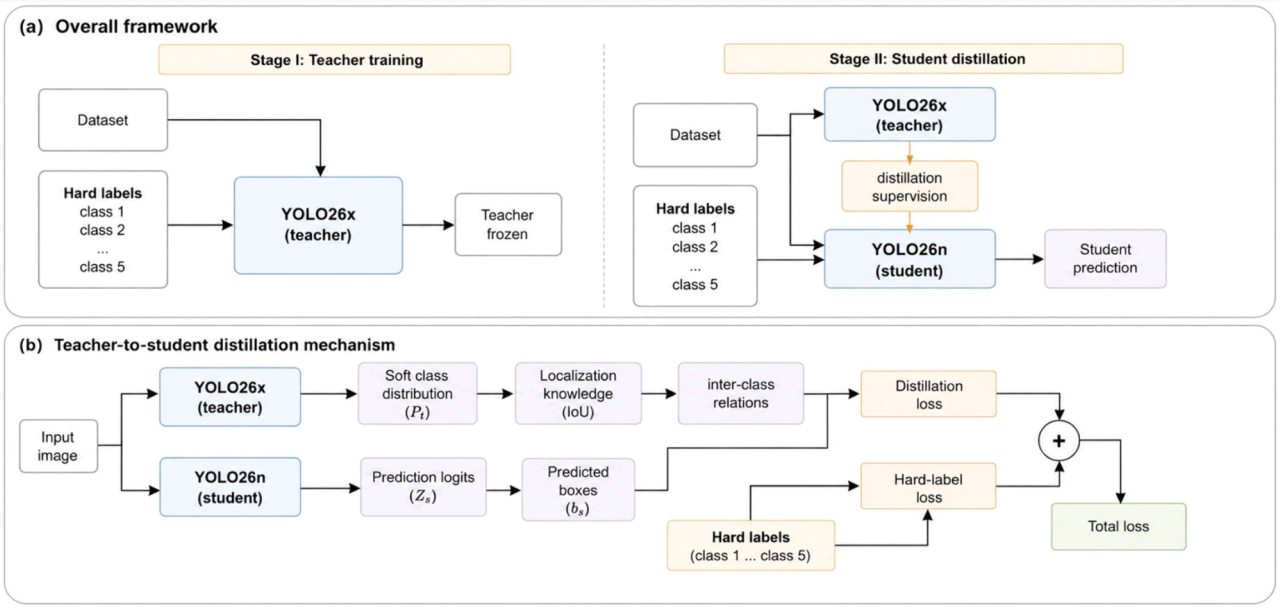

Fig. 7. Framework of the teacher–student knowledge distillation strategy.

supervision information dynamically, without extra generation and storage of independent offline soft labels. The student model is supervised jointly by real hard labels from datasets and real-time distillation loss from the teacher model. This distillation scheme is easy to deploy with no additional storage overhead. It enables the lightweight student model to inherit powerful feature perception capability from the large teacher model, and improves the detection accuracy of blueberry maturity classification while maintaining high inference speed.

2.4. Evaluation metrics

To quantitatively evaluate the comprehensive detection performance of the improved model, this study selects precision, recall, mAP@50, mAP@50:95, the number of parameters, GFLOPs, and FPS as evaluation metrics. Among these, precision, recall, mAP@50, and mAP@50:95 are all obtained based on 5-fold cross-validation with three repeated experiments to ensure the statistical reliability of the results. To maintain an 8:1:1 partition ratio for the training, validation and test sets, one fold is equally split into the validation and test subsets in each run, while the remaining four folds are used as the training set.

Precision and recall are used to assess the degrees of false detection and missed detection of the model respectively. mAP@50 and mAP@50:95 measure the overall detection accuracy under a fixed IoU threshold and multiple gradient IoU thresholds respectively, which can objectively reflect the detection performance of the model in complex field scenarios. The number of parameters and GFLOPs are adopted to evaluate the structural complexity and computational cost of the model, reflecting its lightweight level. FPS refers to the number of images processed per second and characterizes the inference speed of the model. In the FPS test, single-batch input is adopted with an image size of 640 × 640, and the test runs on the designated GPU device with half-precision inference enabled. After 500 warm-up iterations, the latency test is performed with 2000 iterations. The whole test is independently repeated five times with a 30-second interval between each run, and the average value is taken as the final stable FPS result. These indicators can comprehensively compare the performance of different network models from three dimensions: detection accuracy, model scale, and real-time performance.

2.5. Experimental environment

All training and testing experiments of the YOLO26x model were conducted on a high-performance desktop workstation. The workstation is equipped with an Intel Core i9 CPU, 32 GB RAM and a 1 TB solid-state drive, and adopts an NVIDIA RTX 4090 GPU for hardware acceleration. The experimental software environment consists of the PyTorch 2.2.0 deep learning framework and CUDA 11.8 parallel computing platform. Such hardware and software configuration accelerates network convergence and improves the efficiency of model training and image inference.

All hyperparameters in this study are determined through multiple groups of comparative pre-experiments to balance model convergence speed, detection accuracy and training stability. In the basic training phase, the epoch is set to 300, batch size is set to 6, and the number of worker threads is set to 4. Both the initial learning rate and final learning rate are set to 0.01. In the knowledge distillation training phase, the epoch is adjusted to 500, while the initial learning rate and final learning rate are reduced to 0.007. A lower learning rate ensures stable knowledge transfer from the teacher model to the student model during distillation and prevents damage to the original feature extraction capability of the student model.

3. Experimental results and analysis

3.1. Experimenatal results

The proposed YOLO26-EGH model is tested on the self-built greenhouse blueberry maturity dataset, and the category-wise detection results are shown in Table 1. While maintaining a lightweight architecture and satisfactory inference efficiency, the distilled YOLO26-EGH achieves improved overall performance compared with the baseline model, with performance gains of varying magnitudes across all five maturity categories.

The category-specific results demonstrate that the overall performance improvement of YOLO26-EGH does not rely predominantly on any single maturity class. Class-d fully-ripe blueberries possess distinct color and morphological features and therefore achieve relatively high baseline detection accuracy. For the adjacent maturity stages of classes a, b and c, the proposed model yields steady improvements in both mAP@50 and mAP@50:95, which indicates that the introduced

Table 1
Class-wise detection performance and computational efficiency of YOLO26 and YOLO26-EGH on the self-constructed blueberry ripeness dataset.

Model	Class	P (%)	R (%)	mAP@50 (%)	mAP@50:95 (%)	GFLOPs	parameters	FPS
YOLO26	All	87.4	84.9	91.2	79.4	5.2	2375,811	189.5
	a	88.2	84.2	92.3	78.1			
	b	91.1	83.5	89.5	80.4			
	c	84.2	84.4	90.8	80.6			
	d	91.5	92.1	96.2	84.2			
	e	82.1	80.5	87.3	73.5			
YOLO26-EGH	All	87.9	87.9	93.2	82.3	5.7	1841,426	172.4
	a	88.0	85.2	93.7	81.3			
	b	87.2	85.9	92.2	84.6			
	c	89.4	88.1	92.1	82.2			
	d	94.1	95	98.5	89.2			
	e	80.6	85.4	89.6	74.3			

feature-enhancement and multi-scale fusion strategies improve the fine-grained discriminative ability between neighboring maturity grades. Class-e remains the most challenging category for detection, with a relatively small increment in mAP@50:95; nevertheless, its mAP@50 increases from 87.3% to 89.6%, proving that the detection performance for this category is also enhanced.

It should be noted that a slight Precision drop is observed for certain categories such as a, b and e, accompanied by improvements in Recall or mAP metrics, which reflects a certain precision-recall trade-off in the optimized model. Consequently, the performance gains of YOLO26-EGH are consistent across different categories rather than being dominated by a single class.

3.2. Significance test

To quantitatively verify whether the performance improvement of the distilled YOLO26-EGH over YOLO26 is statistically significant and mitigate the influence of randomness from a single training run, repeated experiments combining five-fold cross-validation with three different random seeds were carried out, yielding 15 sets of paired experimental results. The two models adopted identical dataset splits and random seeds in each paired trial. Accordingly, a two-tailed paired-samples t-test was performed to analyze the differences in Precision, Recall, mAP@50 and mAP@50:95. All metrics are presented as mean ± SD, with the number of paired observations ($n = 15$), degrees of freedom ($df=14$), and the significance level set to ($alpha=0.05$). The statistical results are summarized in Table 2.

Statistical analysis shows that Precision only increases slightly by 0.5 percentage points with no significant inter-group difference, indicating that the proposed model maintains stable precision. Recall rises by 3.0 percentage points and reaches a significant level, which demonstrates that the presented method can effectively reduce missed detections and strengthen target-recall capability. mAP@50 and mAP@50:95 increase by 2.0 and 2.9 percentage points respectively, and both improvements pass the significance test, achieving obvious enhancement in the overall detection and localization performance.

Therefore, without a significant fluctuation in Precision, YOLO26-EGH obtains significant improvements in Recall and the two mAP metrics. These statistical results further confirm that the performance gain brought by the proposed improvement strategy is not a random outcome of a single training, and can steadily optimize blueberry maturity detection.

3.3. Ablation studies

To further investigate the individual contributions and synergistic effects of the proposed modules, progressive ablation experiments were conducted on the undistilled model using C2PSA-EMA (Module A), C3k2-GCB (Module B), and the lightweight HSFPN feature pyramid (Module C). The experimental results are presented in Table 3.

The results of single-module ablation experiments reveal that the C2PSA-EMA module can improve the recall of blueberry maturity detection. This attention module enhances the fine-grained feature extraction ability for small targets and alleviates feature loss caused by slight occlusion of branches and leaves in greenhouse scenarios, while leading to a slight drop in inference speed. The C3k2-GCB module can slightly boost recall rate and mean average precision, strengthen the global context capture capability of the network and improve the model adaptability to complex backgrounds, but it increases network parameters and reduces inference speed. The lightweight HSFPN module greatly cuts down computational cost and parameter quantity to realize network lightweight and accelerate inference speed, yet it causes slight declines in detection indicators for blueberry maturity classification.

Multi-module combination experiments verify that these improved modules present obvious complementary and synergistic effects. Although some two-module combinations achieved slightly higher values for individual metrics, the three-module combination provided a more balanced overall performance in terms of detection accuracy, model size, and inference efficiency.This integrated strategy maintains lightweight architecture and rapid inference speed while ensuring competitive detection accuracy, which is more suitable for real-time maturity grading detection of small blueberry targets in actual greenhouse environments.

3.4. Comparative experiments

3.4.1. Ablation comparison of distillation strategy

To verify the comprehensive effectiveness of the proposed YOLO26-EGH improved model and offline knowledge distillation strategy, six groups of comparative experiments are conducted from three dimensions including network backbone scale, network structure optimization and introduction of offline distillation. The experimental

Table 2
Statistical significance analysis of detection performance between YOLO26 and YOLO26-EGH across repeated cross-validation experiments.

Metric	YOLO26		YOLO26-EGH		Improvement (pp)	t	df	p-value	Significance
	mean	std	mean	std					
P (%)	87.4	1.68	87.9	0.87	+0.5	0.459	14	0.6533	ns
R (%)	84.9	1.95	87.9	1.17	+3.0	2.773	14	0.0150	Significant
mAP@50 (%)	91.2	0.44	93.2	0.41	+2.0	6.481	14	1.45×10^{-5}	Significant
mAP@50:95 (%)	79.4	0.40	82.3	0.15	+2.9	13.521	14	1.99×10^{-9}	Significant

Table 3
Ablation study of C2PSA-EMA, C3k2-GCB, and HSFPN modules in the undistilled YOLO26n-EGH model.

Model	Module			P (%)	R (%)	mAP@50 (%)	mAP@50:95 (%)	GFLOPs	parameters	FPS
	A	B	C							
1				86.5	83.3	91.4	78.9	5.2	2375,811	189.5
2	√			86.1	86.5	91.7	78.7	5.2	2420,227	178.8
3		√		87.9	86.9	91.4	78.6	5.2	2326,606	162.6
4			√	87.2	82.8	90.8	77.0	4.6	1647,011	196.6
5	√	√		88.5	85.3	91.8	80.3	6.4	2680,818	146.3
6	√		√	86.9	87.3	91.2	79.6	5.7	1890,631	180.0
7		√	√	86.7	88.9	92.5	80.0	5.7	1797,010	148.9
8	√	√	√	87.2	87.9	92.2	80.8	5.7	1841,426	172.4

results are presented in Table 4.

Horizontal comparison between the original baseline models and EGH improved models with the same backbone demonstrates that the structurally optimized models achieve better detection performance than the original baselines, whether adopting the lightweight YOLO26n backbone or high-precision YOLO26x backbone. It indicates that the proposed network optimization scheme can adapt well to the small-target detection scenario of greenhouse blueberries with slight branch and leaf occlusion and complex background interference, strengthen the model detection capability significantly, and meet the lightweight design requirement of network simultaneously. Vertical comparison between experimental groups with and without offline distillation shows that offline knowledge distillation can further improve model detection performance. This distillation method realizes knowledge transfer relying on high-precision teacher models without modifying the structure of student models. It will not bring extra computational cost and network parameters, nor reduce the inference speed, which can effectively compensate for the performance loss caused by network lightweight optimization.

3.4.2. Comparison of different attention mechanisms

To verify the effectiveness of the EMA attention mechanism adopted in this study, several typical attention mechanisms, including Biformer, CAA, DAT, DLKA, iEAM and SENet, were selected for comparative experiments. The experimental results are shown in Table 5.

Experimental results demonstrate that the model equipped with the EMA attention mechanism achieves the optimal overall performance in blueberry maturity detection. It exhibits superior comprehensive detection accuracy with well-balanced precision and recall, and presents distinct advantages in parameter control and inference efficiency, thus realizing a favorable trade-off between accuracy and lightweight design. Although other attention mechanisms can enhance the feature representation ability of the model in different aspects, among which CAA and DAT show relatively outstanding performance in average precision, and iEAM has certain competitiveness in computational cost and parameter efficiency, none of them can achieve a better comprehensive performance than EMA in terms of detection accuracy, lightweight level and inference speed. This indicates that the EMA attention mechanism is more suitable for the practical requirements of small blueberry target detection in greenhouse environments. It can strengthen fine-grained feature extraction while effectively controlling the computational cost of the model, making it more applicable for the maturity classification and detection task in this study.

3.4.3. Cross-layer synergy performance: C3k2-GCB

To investigate the adaptability and cross-layer synergistic effects of the C3k2-GCB module at different feature levels of the backbone network, layer-wise insertion and multi-layer combination experiments were conducted at the P2, P3, P4, and P5 feature levels using the undistilled model. The experimental results are presented in Table 6.

According to the results of single-layer embedding experiments, embedding the C3k2-GCB module into the P2 layer and P3 layer brings no changes to all detection indicators, which proves that this module fails to enhance feature expression when deployed at shallow network layers and cannot optimize the blueberry maturity detection task. Embedding the module independently into the P4 layer leads to a slight decline in detection performance with negative optimization effects, while embedding it into the high-level P5 layer effectively improves model precision and helps extract global contextual features of small blueberry targets.

The results of multi-layer joint embedding experiments further indicate that any combination containing shallow P2 and P3 layers cannot improve model detection performance. The collaborative embedding of C3k2-GCB modules at P4 and P5 layers achieves the optimal overall detection accuracy. When the module is embedded in all four feature layers simultaneously, the overall model performance does not improve further and remains consistent with that of dual-layer embedding at P4 and P5. It indicates that excessive superposition of C3k2-GCB modules across different layers will cause feature redundancy and fail to further optimize detection performance. Based on the above cross-layer comparative experiments, deploying the C3k2-GCB module only at P4 and P5 layers is reasonable and optimal.

3.4.4. Comparison of different models

To comprehensively evaluate the overall performance and practical advantages of the distilled YOLO26-EGH model for blueberry ripeness detection, comparative experiments were conducted against a range of mainstream lightweight detection models and improved algorithms, including YOLOv8n, YOLOv9t, YOLOv10n, YOLO11n, YOLO12n, the baseline YOLO26n, RT-DETR-L, as well as the improved YOLO11-WBO and YOLO26-WBO models. Among them, YOLO26-WBO was obtained by reproducing and transferring the improvement strategy of YOLO11-WBO to the YOLO26 framework. The experimental results are presented in Table 7.

Table 4
Effects of network scale, structural optimization, and offline knowledge distillation on blueberry ripeness detection performance.

Model	Distillation	P (%)	R (%)	mAP@50 (%)	mAP@50:95 (%)	GFLOPs	parameters	FPS
YOLO26n	No	86.5	83.3	91.4	78.9	5.2	2375,811	189.5
YOLO26n-EGH	No	87.2	87.9	92.2	80.8	5.7	1841,426	172.4
YOLO26x	No	88.0	87.2	91.7	81.9	193.4	55,639,323	93.0
YOLO26x-EGH	No	88.9	89.2	92.6	84.0	162.3	39,862,941	86.7
YOLO26	Yes	87.4	84.9	91.2	79.4	5.2	2375,811	189.5
YOLO26-EGH	Yes	87.9	87.9	93.2	82.3	5.7	1841,426	172.4

Table 5
Detection performance and computational efficiency of different attention mechanisms integrated into the YOLO26n-EGH framework.

Model	P (%)	R (%)	mAP@50 (%)	mAP@50:95 (%)	GFLOPs	parameters	FPS
Biformer	88.3	84.3	91.8	79.2	7.8	2437,635	126.7
CAA	88.3	84.8	92.1	79.5	5.3	2406,659	149.1
DAT	86.4	86.7	92.3	79.5	5.3	2448,643	128.3
DLKA	87.3	86.7	91.8	79.0	5.9	3204,119	90.7
iEAM	87.3	86.4	91.7	79.1	5.2	2388,531	148.8
SENet1	82.0	89.1	91.8	78.8	5.8	2499,350	131.4
EMA	87.2	87.9	92.2	80.8	5.7	1841,426	172.4

Table 6
Effect of C3k2-GCB placement at different P2–P5 feature levels on blueberry ripeness detection performance.

Model	Layer				P (%)	R (%)	mAP@50 (%)	mAP@50:95 (%)
	P2	P3	P4	P5				
1					86.7	88.9	92.5	80.0
2	√				86.6	88.9	92.5	80.0
3		√			86.7	88.9	92.5	80.0
4			√		84.7	87.5	91.8	79.6
5				√	87.7	85.2	92.0	80.3
6	√	√			86.7	88.9	92.5	80.0
7	√		√		84.7	87.5	91.8	79.6
8	√			√	87.9	85.1	92.0	80.3
9		√	√		84.7	87.5	91.8	79.6
10		√		√	87.8	85.2	92.0	80.3
11			√	√	87.2	87.9	92.2	80.8
12	√	√	√		84.7	87.5	91.8	79.6
13	√	√		√	87.9	85.2	92.0	80.3
14	√		√	√	86.3	87.9	92.2	80.3
15		√	√	√	86.3	87.9	92.2	80.4
16	√	√	√	√	86.3	87.9	92.5	80.0

Table 7
Detection performance and computational efficiency comparison of YOLO26-EGH with representative object detection models on the self-constructed blueberry ripeness dataset.

Model	P (%)	R (%)	mAP@50 (%)	mAP@50:95 (%)	GFLOPs	parameters	FPS
YOLOv8n	87.1	86.3	92.3	80.1	8.1	3006,623	269.4
YOLOv9t	88.3	85.8	92.8	80.4	7.6	1971,759	117.5
YOLOv10n	82.9	80.8	89.8	77.1	6.5	2266,143	209
YOLO11n	84.5	85.7	91.6	79.1	6.3	2583,127	216
YOLO12n	85.2	86.9	91.5	78.3	6.3	2557,703	142.7
YOLO26n	86.5	83.3	91.4	78.9	5.2	2375,811	189.5
Rt-detr-l	87.0	82.9	83.3	76.5	103.9	32,009,432	64.2
YOLO11-WBO	86.5	84.5	91.7	80.9	6.6	3554,129	99.8
YOLO26-WBO	91.0	84.7	91.8	80.1	6.8	3452,945	120.7
YOLO26-EGH	87.9	87.9	93.2	82.3	5.7	1841,426	172.4

The Transformer-based RT-Detr-l exhibits mediocre overall detection performance in the fine-grained blueberry maturity detection task. With a large number of parameters and high computational costs, it presents obvious structural redundancy and poor lightweight characteristics. Although mainstream lightweight YOLO series models achieve favorable inference speed, they lack sufficient feature extraction capability for the fine-grained classification of blueberry maturity, resulting in mediocre overall performance in terms of mAP@50 and mAP@50:95. YOLO11-WBO and the reproduced YOLO26-WBO achieve slight improvements in detection performance but at the cost of increased model parameters and computational complexity.

In contrast, the proposed YOLO26-EGH exhibits the optimal comprehensive performance with balanced and outstanding detection accuracy. With a parameter size of 1.84 M, it has the smallest model volume among all comparison models. It maintains a moderate computational load while achieving an inference speed of 172.4 FPS, effectively balancing detection accuracy, lightweight performance, and inference efficiency.

3.4.5. Comparison of different datasets

To verify the generalization performance of the improved model and offline knowledge distillation strategy, comparative experiments are conducted on the public MVBD blueberry dataset captured in complex field environments and the self-built greenhouse blueberry dataset with controllable experimental conditions. The detailed results are presented

Table 8
Cross-dataset generalization performance of YOLO26n, YOLO26n-EGH, and distilled YOLO26-EGH on the self-constructed and MVBD blueberry datasets.

Datasets	Model	P (%)	R (%)	mAP@50 (%)	mAP@50:95 (%)
MVBD	YOLO26n	66.4	73.2	74.3	56.3
	YOLO26n-EGH	68.4	75.1	75.3	57.2
	YOLO26-EGH (Distillation)	78.0	71.1	75.8	58.7
Ours	YOLO26n	86.5	83.3	91.4	78.9
	YOLO26n-EGH	87.2	87.9	92.2	80.8
	YOLO26-EGH (Distillation)	87.9	87.9	93.2	82.3

in Table 8.

Compared with the original YOLO26 baseline model, the proposed YOLO26-EGH model achieves performance gains on both datasets, proving that the improved modules have satisfactory cross-scene generalization ability. All models perform better on the self-built greenhouse dataset than on the MVBD dataset. The main reason is that field scenarios suffer from disordered branch and leaf occlusion, massive background noise and drastic light variation, which make blueberry detection far more challenging than standardized greenhouse environments. After introducing offline knowledge distillation, the model obtains synchronous improvements in precision, recall and mean average precision on the greenhouse dataset with overall optimized detection performance. For the field-based MVBD dataset, the distillation strategy greatly improves model precision with only a minor drop in recall, while the overall mean average precision still increases steadily, and the distilled model outperforms its undistilled counterpart. The comparative results demonstrate that the proposed scheme achieves the best adaptation effect for greenhouse blueberry maturity detection, and it is also capable of adapting to complex field blueberry detection tasks effectively.

3.5. Illumination robustness test

To investigate the robustness of the YOLO26-EGH model against illumination variations, the brightness of test images was scaled from 0.5 to 1.5 at a step of 0.125 to simulate diverse lighting conditions in greenhouse environments. The trained model was directly evaluated on these brightness-adjusted images. The relative errors of precision, recall, mAP@50 and mAP@50:95 were calculated. The relative error is defined as the difference between the metric value after transformation and the original value, divided by the original metric value. The experimental results are shown in Fig. 8.

Under low-light conditions where brightness is lower than the original images, all detection metrics of the model only decline slightly, and the overall performance remains stable. When the brightness multiplier exceeds 1.25, corresponding to strong illumination, obvious degradation can be observed for all metrics, with the most severe performance drop occurring at a brightness multiplier of 1.5. In practical greenhouse applications, such extreme strong light conditions rarely occur, which generally only appear at noon without shading when fruits are directly exposed to intense sunlight. Overall, the model exhibits favorable detection robustness within the common illumination fluctuation range of greenhouses, and its performance is significantly affected only under extreme strong light.

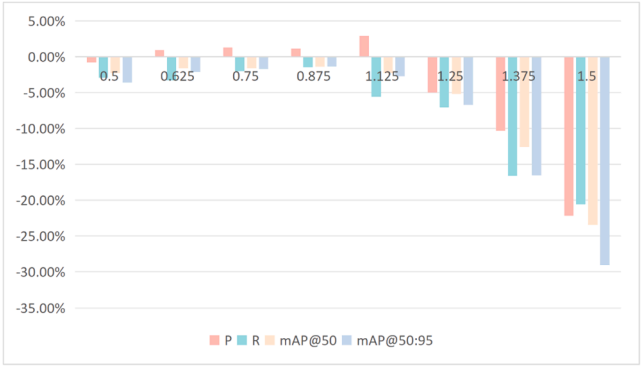

Fig. 8. Relative error of detection metrics under different brightness multipliers.

4. Discussion

4.1. Analysis of model classification error

The normalized confusion matrix illustrates the classification performance of the baseline YOLO26n and the proposed YOLO26-EGH models for blueberry maturity levels based on the self-built greenhouse dataset and the public MVBD field dataset. Compared with YOLO26n, the YOLO26-EGH model alleviates maturity misjudgment and background interference to a certain degree, presenting slightly improved classification stability. Recognition errors across the two datasets exhibit consistent characteristics. Misclassification mainly occurs between adjacent maturity levels while cross-level confusion is extremely limited, which demonstrates the capability of the models to capture continuous ripening features of blueberries, the result as shown in Fig. 9.

On the self-built greenhouse dataset, all maturity classes achieve competitive diagonal classification accuracy. Class d corresponding to fully mature blueberries yields the highest accuracy of 0.93. The accuracy values of class a and class b are both 0.89, and those of class c and class e are 0.85 and 0.90, respectively. These results verify the reliable classification performance of the model for blueberry samples at different developmental stages.

The main limitation lies in the separation between targets and background. Small immature fruits in Class a are easily misclassified as background with a misclassification rate of 0.36, and late expansion stage fruits in Class e also show similar misclassification toward the background. A small number of background samples are also mistakenly detected as fruits. This issue arises from the high similarity in texture features between small and weak fruits and greenhouse branches and leaves, which represents a key challenge in current feature extraction. Subsequent optimization can focus on strengthening the differential feature representation between small-target fruits and background interference. The misclassification of background as fruits is not caused by the model mistaking non-target objects such as branches and leaves for fruits. Instead, it occurs because incomplete fruits at image edges and small, distant blurred fruits are not included in the annotation, and the detection of such unlabeled targets by the model results in statistical false detections.

Further analysis shows that, cross-level misclassification is mainly affected by complex illumination conditions. Lighting variations weaken the visual feature differences between blueberries at different maturity stages, leading to a small amount of cross-level confusion. Misclassification between adjacent maturity classes results from the continuous and gradual ripening process of blueberries. Some fruits are in the transition stage between two maturity levels and exhibit intermediate feature properties, making them more prone to classification deviations. In practical deployment applications, manual post-processing can be used to correct such transitional samples by uniformly assigning them to the previous class or the adjacent higher class, so as to improve the stability and consistency of actual grading results.

Model performance on the public MVBD field dataset is slightly inferior to that on the greenhouse dataset with more frequent cross-level confusion. Variable field illumination and complex environmental conditions homogenize the visual features of blueberries at different maturity stages, increasing classification difficulty. Furthermore, semi-ripe transitional samples of class b and class c are insufficient, with only one-tenth of the sample quantity of class a. Such unbalanced data distribution weakens the model's recognition ability for transitional maturity features and increases category confusion.

4.2. Visual comparison of detection performance between baseline and improved models

Three typical images were selected from both the self-built greenhouse blueberry dataset and the public MVBD field dataset for visual

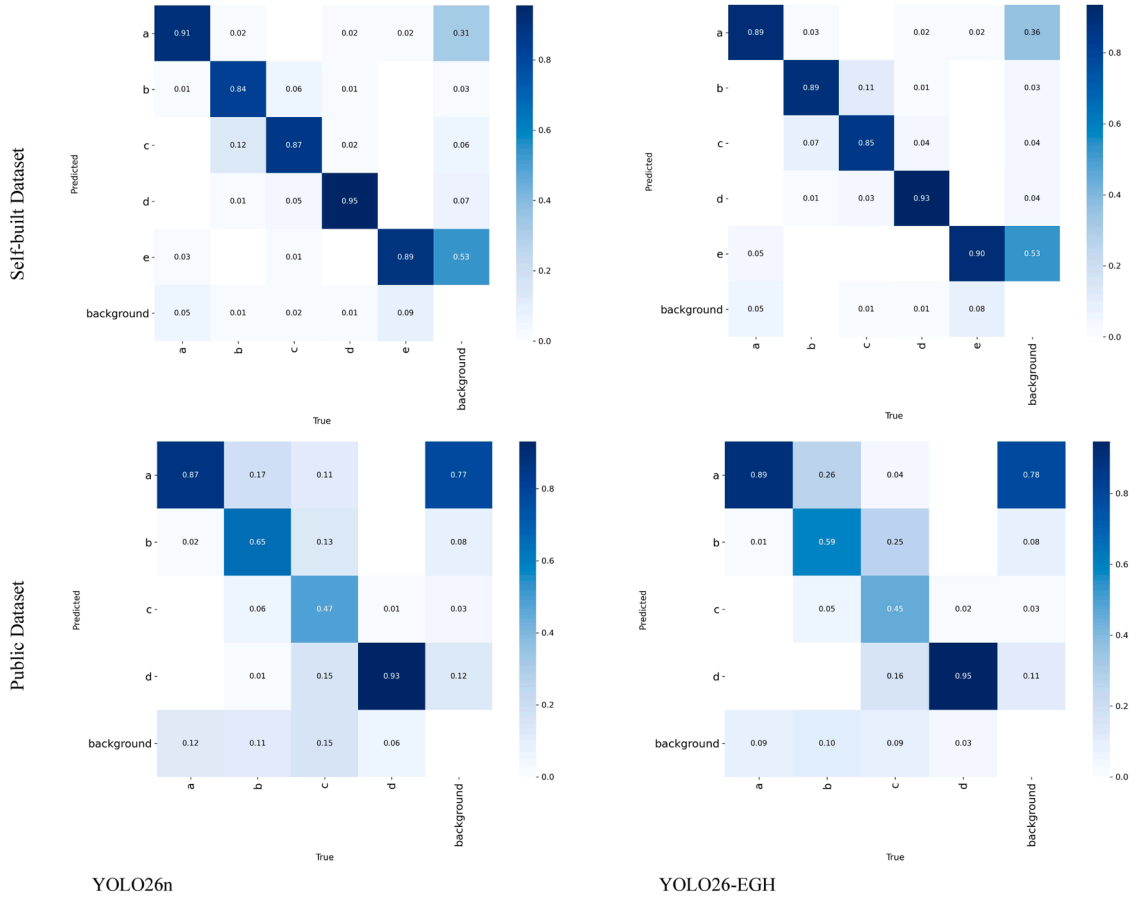

Fig. 9. Normalized confusion matrix.

comparison of detection results. The original YOLO26n, the undistilled YOLO26-EGH, and the distilled YOLO26-EGH (Distillation) were used for qualitative analysis of detection performance differences, as shown in the Fig. 10.

On the self-built dataset, all models can effectively detect targets in the first group of images, among which the distilled YOLO26-EGH achieves higher detection confidence. In the second group of images, the early coloring-stage blueberry at the lower-right corner is missed by the undistilled YOLO26-EGH due to blurred imaging, while YOLO26n yields misclassification due to target overlap interference. In the third group of images, the mature blueberry at the lower-left corner is incorrectly identified as late half-mature by both the undistilled YOLO26-EGH and YOLO26n under local overexposure. On the public field dataset, the distilled YOLO26-EGH also exhibits higher detection confidence and more reliable classification accuracy, with overall detection robustness superior to the undistilled model and the baseline model.

4.3. Comparative analysis of feature attention regions using heat maps

In this section, the XGrad-CAM heat map visualization method is employed to intuitively compare the feature activation distributions in the small-target feature layers before the detection head among the baseline YOLO26n, the undistilled YOLO26-EGH, and the distilled YOLO26-EGH models. Verification is conducted on two typical scenarios in greenhouse environments: dense fruit clusters and leaf-occluded targets. The feature heat map results are shown in the Fig. 11.

It can be observed from the heat maps that the baseline YOLO26n exhibits relatively dispersed feature activation on fruit regions, with certain responses to background elements such as leaves and branches, leading to weak feature focusing in cases of dense clusters and occlusions. The improved YOLO26-EGH shows enhanced concentration of feature activation on target regions and alleviated redundant background activation, presenting more stable target feature extraction capability. The YOLO26-EGH integrated with offline knowledge distillation achieves more compact feature activation, stronger responses to core fruit areas, and further reduced background interference. The overall discriminability of features is improved, which contributes to its superior performance in detection confidence and classification accuracy.

4.4. Deployment analysis of model performance

To verify the edge deployment performance and engineering application value of the proposed algorithm, this paper conducts embedded edge deployment experiments on the YOLO26-EGH model. The model is trained on the Windows platform, with a parameter count of 1841,426 and a computational cost of only 5.7 GFLOPs. The inference latency on the PC side reaches 5.80 ms, showing the advantages of lightweight design and high inference speed, which meets the basic requirements for edge deployment.

This study selects the NVIDIA Jetson Orin Nano SUPER with 8 GB memory as the edge deployment hardware, as shown in Fig. 12. The hardware device and detection results are shown in the figure. After format adaptation and inference optimization of the trained YOLO26-EGH model, actual tests are carried out on the embedded platform. The results show that the inference latency for single-image detection is 8.68 ms, the latency for real-time detection is 12.72 ms, and the average single-frame inference time is 10.3 ms.

Limited by the constrained computing power and storage resources of embedded devices, the inference speed of the model decreases to a certain extent compared with the PC platform, which is a normal

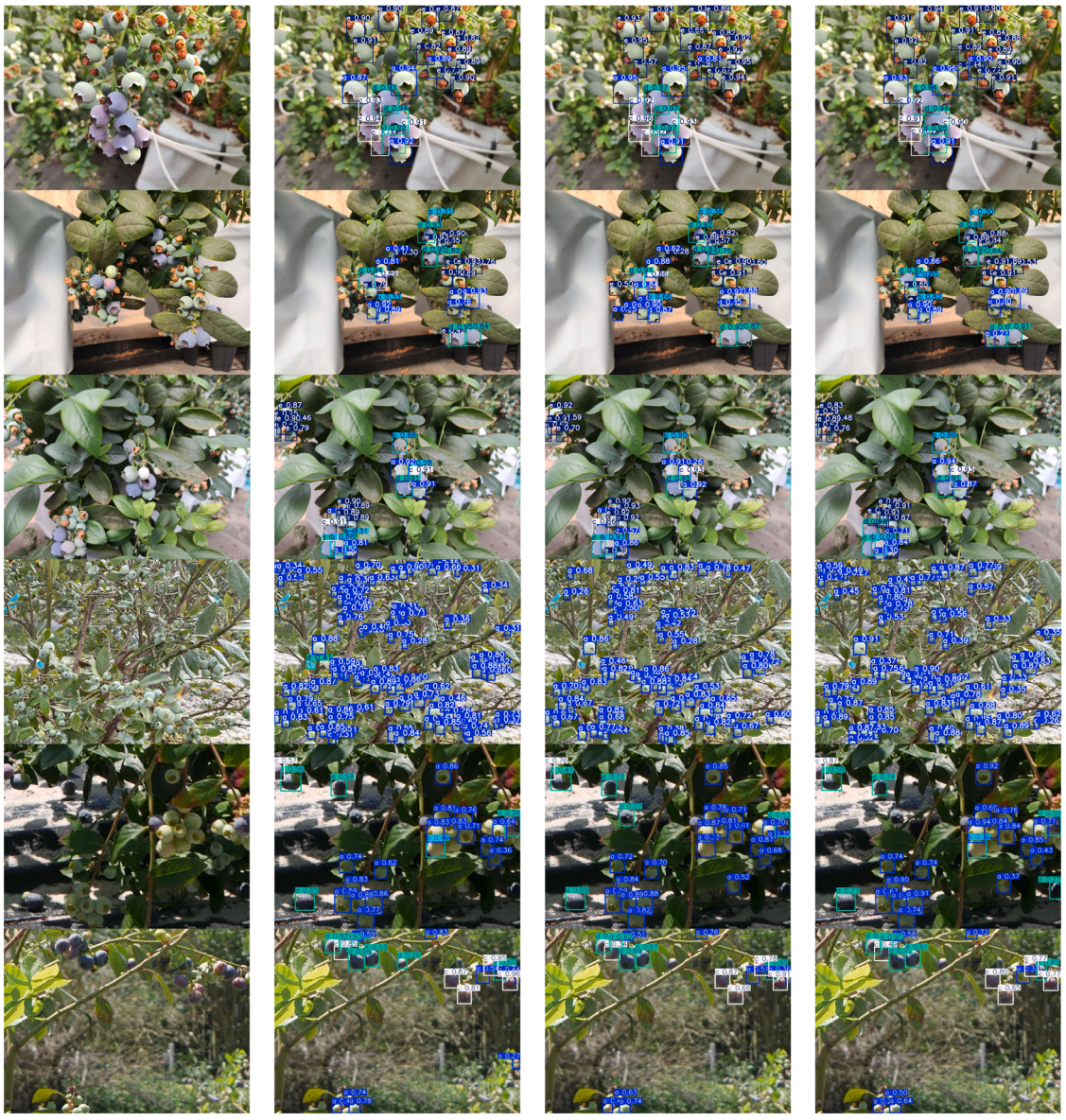

Fig. 10. Visual comparison of YOLO26-EGH under different scenarios.

hardware performance difference. The proposed YOLO26-EGH model can be well adapted to embedded edge devices. While ensuring the detection accuracy of blueberry maturity, it also demonstrates feasibility for engineering deployment and can meet the application requirements of field intelligent monitoring equipment for blueberries.

4.5. Limitations and future research directions

This study optimizes the YOLO26 network structure and adopts knowledge distillation strategy for blueberry maturity detection, and comparative experiments are conducted based on both self-built greenhouse datasets and public field datasets. The proposed method achieves preferable detection and classification performance under leaf occlusion, local overexposure and dense fruit stacking scenarios. Nevertheless, there are still several limitations in current research. First, all experimental data are collected under conventional natural illumination conditions, and images captured under extreme environments such as strong backlight, heavy shadow, rainy and foggy weather are not included. Second, the robustness of the model under extreme field environments needs to be further improved. Secondly, this research only focuses on single-frame static image detection, without combining the growth timing characteristics of blueberries. The model cannot use continuous growth state information to assist maturity classification. Third, for extremely difficult samples with full leaf occlusion and tight overlapping fruits, the model still produces minor classification errors, and the fine-grained distinguishing ability for adjacent maturity levels can be further enhanced. Finally, restricted by the blueberry growth cycle, the number of fruits in the field was low and immature during the field deployment test. This study only verified the inference speed on edge devices, and did not fully evaluate the maturity classification accuracy in real field scenes. Therefore, the comprehensive performance of the model in edge deployment still needs to be further improved.

In view of the above limitations, further research will be carried out from three aspects in the future. First, expand the dataset by supplementing blueberry images captured under extreme illumination and bad weather conditions, so as to enhance the environmental adaptability and overall robustness of the detection model. Second, combine the growth timing characteristics of blueberry fruits to explore timing auxiliary

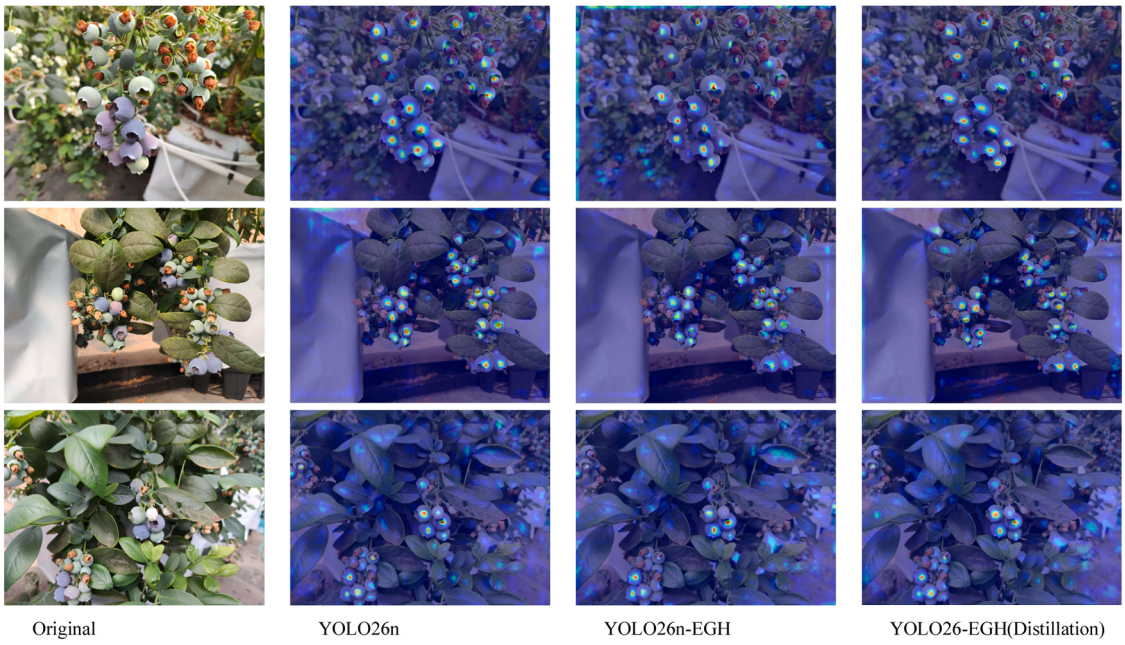

Fig. 11. Heatmap comparison of YOLO26-EGH under different scenarios.

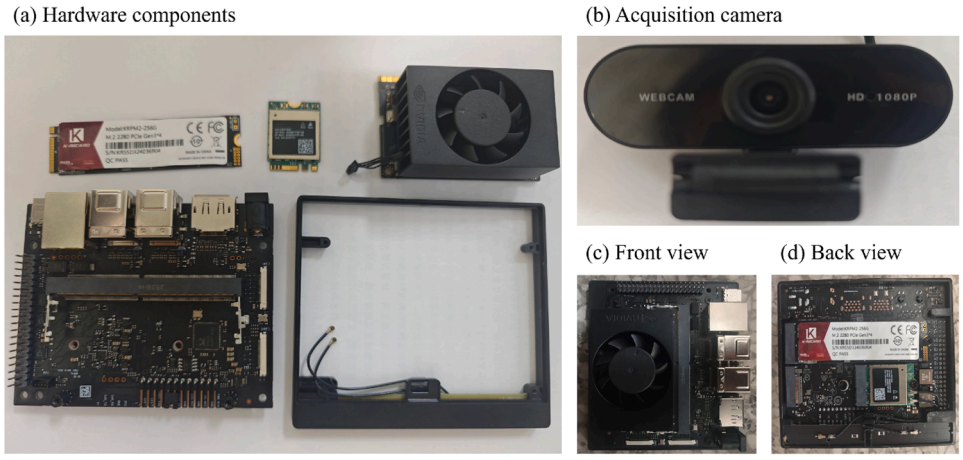

Fig. 12. Edge deployment device.

classification strategies, and reduce misclassification rates between adjacent maturity stages. Third, introduce fine-grained feature enhancement modules to further suppress background interference, optimize the feature extraction effect of severely occluded small target fruits, and promote the practical application of the detection algorithm in intelligent blueberry field monitoring systems. Finally, a complete field edge deployment experiment will be carried out during the blueberry full-fruiting and ripening period to simultaneously verify the detection accuracy and real-time performance of the model in real field environments, further promoting the practical application and engineering iteration of the algorithm in blueberry intelligent monitoring systems.

5. Summary

This study proposed YOLO26-EGH, a lightweight knowledge-distilled detection model for blueberry ripeness grading under greenhouse conditions. A two-year blueberry image dataset containing F6 and F11 cultivars was constructed, and 3000 valid images were annotated into five maturity grades to support fine-grained detection research. By integrating C2PSA-EMA, C3k2-GCB, HSFPN and a teacher–student knowledge distillation strategy, the model enhanced the extraction of subtle color, texture and contextual features while reducing structural redundancy.

Quantitative experimental results show that the proposed model achieves a compact parameter size of 1.84 M and a low computational cost of 5.7 GFLOPs, along with a desktop inference speed of 172.4 FPS. It yields mAP@50 and mAP@50:95 values of 93.2% and 82.3%, respectively, achieving an effective balance between model lightweightness and detection accuracy compared with the baseline model. Per-class evaluation reveals that the model maintains stable recognition performance for most mature blueberry samples, while relatively lower accuracy is observed for immature fruits at the late expansion stage with residual floral structures. In addition, transitional fruits with ambiguous maturity characteristics easily cause minor classification deviations between adjacent maturity levels.

Ablation experiments validate the synergistic effectiveness of the three designed modules. Comparative experiments on attention mechanisms and feature layer deployment further verify that the EMA mechanism combined with feature enhancement at the P4 and P5 layers

yields the optimal detection performance. Illumination robustness tests demonstrate that the model maintains stable adaptability under ordinary light fluctuations, whereas its performance degrades under extreme strong-light interference. Cross-domain validation on the public MVBD dataset confirms the superior generalization capability of the proposed model in both greenhouse and complex field scenarios.

Confusion matrix visualization, detection comparison, and Grad-CAM heatmap analysis systematically reveal the error sources, including gradual maturity feature transition, small-target blurring, leaf occlusion, and complex environmental noise. Meanwhile, the structural modifications and knowledge distillation strategy adopted in this study effectively suppress redundant background feature activation, strengthen feature focusing on fruit regions, and mitigate false and missing detections, thereby significantly improving the detection stability and classification accuracy in complex orchard environments.

Embedded deployment tests demonstrate that the YOLO26-EGH model achieves stable single-frame inference of 10.3 ms on the NVIDIA Jetson Orin Nano SUPER device, meeting the engineering requirements for real-time field monitoring.Overall, the YOLO26-EGH model proposed in this paper can effectively improve the accuracy and stability of blueberry maturity grading detection in complex environments, and possesses the potential for real-time edge deployment. It can provide technical support for precise blueberry harvesting and intelligent orchard monitoring equipment.

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CRediT authorship contribution statement

Huiqin Li: Writing – original draft, Validation, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Bozhou You:** Writing – original draft, Visualization, Formal analysis, Conceptualization. **Youyang Cheng:** Writing – review & editing, Validation, Data curation. **Jinfeng Peng:** Supervision, Project administration, Investigation. **Enyu He:** Investigation, Conceptualization. **Fuzhong Li:** Resources, Funding acquisition. **Xiaoying Zhang:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influ ence the work reported in this paper.

Data availability

Data will be made available on request.

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